

TREEMMM: Tree-Based Marketing Mix Modeling with SHAP Attribution and Automatic Interaction Discovery

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Abstract

Marketing mix modeling (MMM) at the healthcare-professional (HCP) panel scale presents a regime fundamentally different from the aggregate weekly time-series that motivated classical Bayesian MMM. We present TREEMMM, a Python package pairing LightGBM with SHAP for distribution-aware channel attribution, automatic interaction discovery, and marginal-ROI optimization on panel data. We benchmark TREEMMM against five baselines (GLMM-Naive, GLMM-Oracle, PyMC-Hier-Naive, PyMC-Hier-Oracle, PyMC-Marketing) on four synthetic datasets. On the three non-linear datasets, TREEMMM achieves attribution share-MAPE of $17.9\% \pm 0.2\%$ vs. $22.2\% \pm 0.3\%$ for GLMM-Naive (gap 4.3 pp ± 0.4 pp, $N=5$ seeds). A distributional-GLM experiment confirms the advantage is not an artifact of the $\log_{1p}$ link workaround: a properly specified Poisson/Tweedie/Gamma GLM (GLMMDist-Naive) performs no better, and on two of three datasets slightly worse. TREEMMM automatically detects 5 of 6 planted channel interactions without pre-specification ($F1 = 0.56$ at default thresholds); regression-based baselines detect zero. On a geo-panel DGP designed for PyMC-

Marketing’s parametric strengths, TREEMMM(29.7%) outperforms PyMC-Marketing (52.1%) and Meridian (57.0%). A prior-variance sweep confirms that at $n=3,000 \times 36$ periods all channels move less than 0.1 pp across a $4 \times$ prior-width bracket. Code and data: <https://github.com/jamesyoung93/treemmm>.

Keywords: marketing mix modeling, SHAP attribution, gradient boosted trees, heterogeneous treatment effects, panel data, interaction discovery, LightGBM

JEL codes: C14, C23, M31, C53

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1 Motivation and Scope

1.1 The “Bayesian MMM is superior” claim is regime-conditional

The conventional framing that Bayesian methods are the right default for marketing mix modeling was forged in a specific data regime: aggregate weekly time-series with roughly 150 observations, ten or more collinear channels, and stakeholder demand for posterior intervals. In that regime the conventional framing is correct. Frequentist ordinary least squares struggles with weakly identified adstock and saturation parameters, sign-flipping is a real symptom, and priors regularize estimation while allowing informed transfer of prior experimental knowledge. These advantages are real, and they are real because of the data scarcity and aggregation that define the classical MMM problem.

What changes the calculus is that modern pharmaceutical and B2B promotional analytics rarely produce data in that regime. Promotional data at the healthcare-professional (HCP) panel level (10,000 HCPs and 150 weeks gives roughly 1.5 million observations) sits in a different statistical environment. Four panel-data shifts move the right answer away from “default to Bayesian.”

Cross-sectional variation breaks multicollinearity. With 1.5 million observations spanning HCPs and time, the rank deficiency that plagues aggregate weekly MMM disappears. The dominant identification path is now within-stratum cross-sectional contrast, not the national-trend separation Bayesian priors are designed to regularize.

Business constraints bound counterfactuals to in-support regions. Minimum and maximum touch constraints per HCP force any budget reallocation to redistribute over the existing HCP universe rather than push any individual into unseen territory. Counterfactuals stay within the convex hull of training support, which is exactly where tree-based models are reliable.

Composite potential variables absorb observable selection bias. Including a well-constructed potential composite (historical prescribing, panel size, specialty, practice type, payer mix, market share trends) means that within strata of potential, variation in promotional intensity approaches as-if-random. Conditioning on this composite implicitly approximates what double machine learning and propensity-score methods do formally.

Out-of-sample validation becomes feasible. With millions of panel observations, proper time-series cross-validation produces real out-of-sample prediction guarantees. Aggregate Bayesian MMM with 150 observations has no honest holdout large enough to validate response-curve shape.

Under these four shifts, tree-based methods with SHAP attribution are competitive with or superior to Bayesian MMM for most promotional decisions. Under the classical aggregate regime, they are not.

1.2 What TREEMMM is and is not designed for

TREEMMM is designed for HCP-level (or store-level, account-level, or geo-cell-level) panel data with meaningful cross-sectional variation independent of temporal variation. It is designed for settings where business constraints bound counterfactuals to in-support regions, where observable confounders dominate (rich covariates absorb selection bias), and where the deliverable is response-shape recovery and channel ranking rather than probabilistic statements about a single scalar. It needs sample sizes that support proper time-series cross-validation—in practice at least a few thousand panel-period rows.

TREEMMM is *not* designed for, and this paper does not claim superiority in, several other regimes.

- *Aggregate weekly time-series with low observation count* (≈ 150 observations across ten or more channels) is Bayesian MMM territory. TREEMMM has fewer degrees of freedom to spare and no native adstock or saturation parameterization.

- *Brand-new channels* with no historical variation cannot be modeled by a tree. A tree cannot estimate the effect of an input it has never seen at any non-zero level.
- *Formal probabilistic statements* with parameter-level credible intervals are outside TREEMMM’s scope. It provides prediction intervals via conformalized quantile regression and bootstrap, not parametric posterior intervals [Romano et al., 2019].
- *Hierarchical sparse-cell estimation* is a poor fit. Where the structure of interest has many small cells (e.g., 200 HCPs by 50 specialties by 30 payers, with most cells holding fewer than five observations), partial pooling via a Bayesian hierarchical model dominates.

1.3 Decision branches

The paper’s headline claims live on a concrete branch of the decision tree. The branch favors Bayesian structural MMM when most of the following hold: an observation-to-parameter ratio below 20, a design-matrix condition number above 30, treatment variation primarily temporal with channels co-moving (correlation above 0.7), credible informative priors from external evidence, decisions that require probabilistic statements, and hierarchical units with sparse cells. The branch favors a tree-based approach (TREEMMM) when most of the following hold: panel structure with cross-sectional variation independent of temporal, treatment overlap reasonable across covariate strata, counterfactuals that stay within empirical support, confounders that are largely observable, nonlinearities and interactions that dominate the response surface, prediction accuracy and out-of-sample validation as deliverables, and a sample size that supports proper cross-validation.

The four DGPs in the paper (pharma_brand, cpg_brand, saas_brand, linear_baseline) are designed to live on the right-hand branch. The linear DGP is included as an honesty test: when the data-generating process is linear and Gaussian, which is the natural home turf of a GLMM or Bayesian regression, TREEMMM should not dominate. The benchmark

in [Section 3](#) confirms this. On the linear DGP, TREEMMM posts $0.4\% \pm 0.1\%$ attribution-share MAPE against GLMM-Naive’s $0.3\% \pm 0.0\%$.

1.4 Risks of misuse

The paper does not argue that tree-based methods are categorically safer than Bayesian methods. They have different failure modes.

Bayesian-specific failure modes include tight priors centered on desired conclusions, identifiability problems masked by smooth posteriors, MCMC convergence theater, false precision from credible intervals on misspecified models, and false confidence in extrapolation along parametric forms. Tree-specific failure modes include confident-looking SHAP attributions in collinear settings without sensitivity checks, flat extrapolation outside support, and the temptation to treat prediction accuracy as causal validity. Shared failure modes include causal claims from observational data without an identification strategy.

Defensible practice in either paradigm requires showing sensitivity, support, and identifiability for the recovered effects. [Section 5](#) documents which of these checks this paper has executed.

2 Methods

2.1 Distribution-Aware Objective Selection

TREEMMM supports four objectives matched to outcome distributions:

Table 1: Supported objectives and their use cases.

Distribution	Objective	When to Use
Gaussian	MSE	Continuous, symmetric (revenue, value sales)
Poisson	Log-link	Non-negative counts (Rx, orders, NPS)
Tweedie	Log-link	Zero-inflated continuous (revenue with stockouts)
Gamma	Log-link	Strictly positive continuous (per-transaction revenue)

An automated diagnostic examines discreteness, zero-inflation rate, skewness, and the mean-variance relationship to recommend an objective. The user can override this recommendation.

2.2 Link-Function-Aware Attribution Decomposition

SHAP TreeExplainer computes values in margin space. For identity-link (Gaussian) models, SHAP values are directly additive on the response scale. For log-link (Poisson/Tweedie/Gamma) models, naive exponentiation breaks additivity: $\exp(a + b) \neq \exp(a) + \exp(b)$. TREEMMM’s decomposer uses unsigned proportional allocation:

$$\hat{a}_i = \frac{|\phi_i|}{|\phi_0| + \sum_j |\phi_j|} \cdot \hat{y} \quad (1)$$

where ϕ_i is the SHAP value for feature i , ϕ_0 is the base value, and $\hat{y}$ is the model prediction. This guarantees (1) non-negative attributions (standard MMM practice), (2) attributions sum to the predicted outcome for every observation, and (3) the relative ranking of channels is preserved from the log-space SHAP values.

2.3 Temporal Cross-Validation

TREEMMM uses time-respecting validation to prevent future data leakage: rolling-origin training windows grow forward, and test windows are always future periods. The minimum training fraction is configurable (default: 50% of periods). No observation from a future period ever appears in the training set.

2.4 Hyperparameter Tuning

Optuna Bayesian optimization tunes tree hyperparameters within each cross-validation fold: number of leaves, learning rate, regularization (L1/L2), feature/bagging fraction, and minimum child samples. The tuning objective is distribution-matched (Poisson deviance for counts, Tweedie deviance for zero-inflated outcomes, MSE for Gaussian).

2.5 Marginal ROI Simulation with Extrapolation Safety

The mROI simulator estimates response curves with a critical safety constraint: per-customer caps are set at the observed percentile (default: 95th), and higher aggregate engagement is achieved by spreading to more customers rather than pushing any individual beyond observed bounds. The constrained optimizer (scipy SLSQP) surfaces which customers have untapped marginal sensitivity, revealing that “see your top customers the most” is often suboptimal.

2.6 Baseline Models

We compare TREEMMM against five baselines spanning three modeling paradigms.

GLMM-Naive. Main effects only, random intercepts per customer (statsmodels MixedLM). Represents a typical analyst who does not specify interactions or distributional families.

GLMM-Oracle. Correctly specified interaction terms matching the DGP (statsmodels MixedLM). Represents an analyst with perfect knowledge of which variables interact.

Both GLMM configurations use a log-transform workaround on non-Gaussian datasets (log-transform the outcome before fitting MixedLM), which approximates a log-normal model. This is a common practitioner workaround. A properly specified distributional GLMM (e.g., `glmmTMB` in R) is not equivalent; see [Section 6](#) for discussion.

PyMC-Hier-Naive. A hierarchical Bayesian linear MMM fit on the full panel (3,000 entities $\times$ 36 periods, $\approx 108,000$ rows), with main effects only and a customer-level random intercept. Prior structure (after standardization): $\alpha \sim \mathcal{N}(0, 5)$, $a_{\text{cust}} \sim \mathcal{N}(0, \sigma_{\text{cust}})$ with $\sigma_{\text{cust}} \sim \text{HalfNormal}(2)$, $\beta_{\text{main}} \sim \mathcal{N}(0, 1)$, and $\sigma_{\text{obs}} \sim \text{HalfNormal}(1)$. Posterior sampling uses `nutpie` (Rust-backed NUTS) with four chains, 1,000 tuning and 1,000 draw steps each.

PyMC-Hier-Oracle. Identical to PyMC-Hier-Naive but with the planted DGP interaction terms added as fixed effects ($\beta_{\text{inter}} \sim \mathcal{N}(0, 0.5)$). Forms the Bayesian counterpart of GLMM-Oracle.

PyMC-Marketing. The leading open-source Bayesian MMM [[PyMC Labs, 2025](#)], implementing NUTS sampling with geometric adstock and logistic saturation transforms on aggregate time-series (one row per time period, 36 rows). We use default weakly informative priors, `GeometricAdstock(l_max=4)`, `LogisticSaturation()`, and the NumPyro sampler (500 draws, 500 tuning, 2 chains).

[Table 2](#) summarizes the three PyMC variants on axes that matter for interpreting results.

Table 2: Summary of PyMC model variants used in the benchmark.

Axis	PyMC-Hier-		PyMC-Hier-		PyMC-	
	Naive		Oracle		Marketing	
Data shape	Panel	(108K	Panel	(108K	Aggregate	(36
	rows)		rows)		rows)	
Random ef-	Random	inter-	Random	inter-	None	
fects	cept		cept			
Fixed effects	Main effects		+ DGP interac-		Main effects	
			tions			
Adstock	None		None		Geometric	
Saturation	None		None		Logistic	
Headline non-	$22.7\% \pm 0.3\%$		$18.1\% \pm 0.3\%$		$75.1\% \pm 5.5\%$	
linear MAPE						

2.7 Experimental Design

2.7.1 Synthetic Datasets

We evaluate on four synthetic datasets with known ground-truth DGPs (Table 3). All non-linear datasets use heterogeneous customer sensitivity (HCS), where each customer draws a latent sensitivity vector from a segment-specific multivariate normal distribution. Each dataset uses $3,000 \text{ entities} \times 36 \text{ periods}$.

Table 3: Synthetic dataset specifications.

Dataset	N	Distribution	Channels	Non-lin	Interactions	HCS
Pharma	$3K \times 36$	NegBin ($r=5$)	6	log, sqrt, lin	3	2 seg
CPG	$3K \times 36$	Tweedie ($p=1.5$)	5	sqrt, log, lin	1	3 seg
SaaS	$3K \times 36$	ZI-Gamma (ZI=10%)	5	sqrt, log	2	2 seg
Linear	$3K \times 36$	Gaussian	3	None	None	None

The pharma DGP (illustrative functional form):

$$\lambda_{i,t} = \exp\left(\beta_0 + \sum_k \beta_{i,k} f_k(x_{i,k,t}) + \gamma_1 x_{\text{rep},i,t} x_{\text{samples},i,t} + \gamma_2 x_{\text{dtc},i,t} x_{\text{rep},i,t} + \gamma_3 x_{\text{peer},i,t} x_{\text{rep},i,t} + \delta Z_{i,t}\right), \quad y_{i,t} \sim \text{NegBin}(\lambda_{i,t}) \quad (2)$$

where f_k is a channel-specific response function (log, sqrt, or linear), $\beta_{i,k}$ is drawn from a segment-specific multivariate normal (heterogeneous customer sensitivity), and γ controls interaction strength.

2.7.2 Evaluation Metrics

The evaluation has two distinct axes that should not be conflated. *Predictive accuracy* asks how close outcome predictions are to held-out values, summarized by R^2 and weighted MAPE on response-scale predictions. *Attribution recovery* asks whether the model’s decomposition of outcomes onto channels matches the true DGP, summarized by:

1. **Attribution share-MAPE:** mean absolute percentage error between recovered and reference attribution shares (only for variables with reference share $> 0.5\%$);
2. **Rank correlation:** Spearman ρ between recovered and true attribution rankings;
3. **Interaction detection:** whether both variables in a planted interaction exceed 3% SHAP importance;

4. **HCS recovery:** Spearman ρ between true latent customer sensitivity and customer-level mean $|\phi|$;
5. **Distribution matching:** whether the correctly matched objective outperforms the mismatched objective.

The headline metric in [Section 3](#) is attribution share-MAPE, not predictive MAPE on responses. Reference attribution shares are computed as the L_1 -norm of mean-centered DGP component contributions, normalized to sum to one. This is a variance-attribution heuristic, not the Shapley decomposition of the DGP function; a Monte Carlo Shapley ground truth is planned for a future release.

2.7.3 Benchmark Configuration

TREEMMM is trained with LightGBM using 20 Optuna hyperparameter trials per fold, searching over learning rate, number of leaves, minimum child samples, L1/L2 regularization, and feature/bagging fractions. Maximum tree depth is constrained to $[3, 5]$ and monotone constraints are enabled for all promotional variables (non-negative effect direction). Attribution MAPE is computed by comparing L_1 -centered SHAP-derived channel shares against L_1 -centered ground-truth shares, both on the margin (log/link) scale, which is methodologically consistent with SHAP’s native decomposition on the model’s link function.

3 Benchmark Results

All results are mean $\pm$ SE across $N=5$ seeds (seeds 0–4) unless stated otherwise.

3.1 Attribution Recovery

Figure 1: Attribution Recovery Across 4 Benchmark Datasets

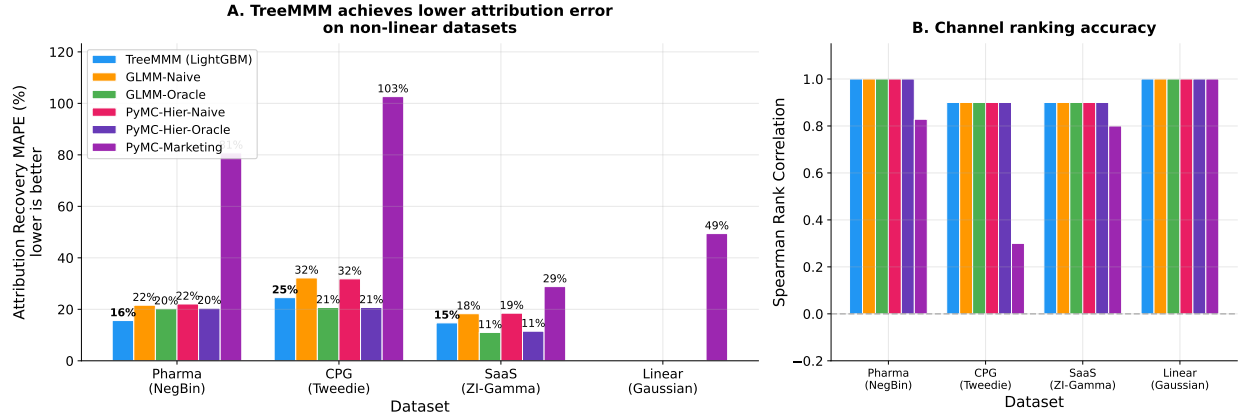


Figure 1: Attribution recovery across four benchmark DGPs. Left: attribution share-MAPE (lower is better). TREEMMM (blue) achieves lower share-MAPE than GLMM-Naive (orange) on all three non-linear datasets. Right: Spearman rank correlation.

Figure 2: Ground Truth vs. Recovered Attribution Shares
TreeMMM correctly identifies relative channel importance across datasets

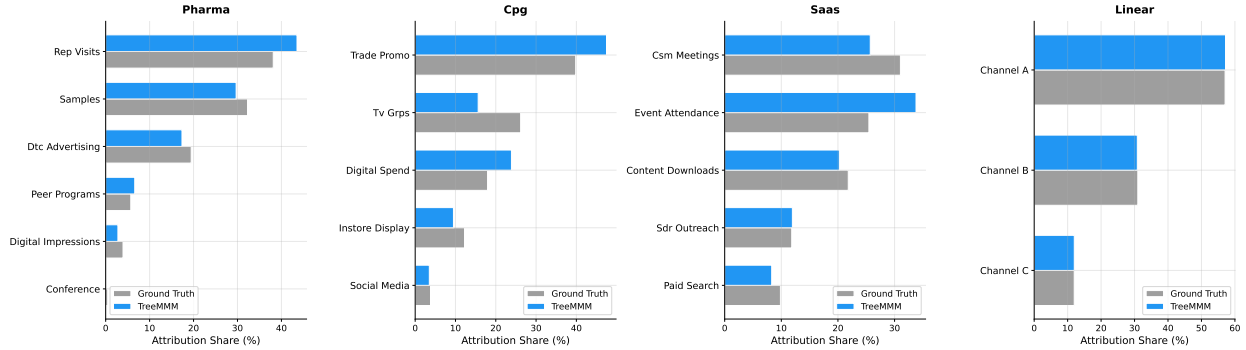


Figure 2: Attribution shares per channel. Reference shares (gray) versus TREEMMM-recovered shares (blue) for each channel within each dataset.

Table 4 shows attribution recovery across all four datasets and eight models. Results are from the full-scale benchmark (3,000 entities $\times$ 36 periods). All numbers are mean $\pm$ SE across $N=5$ seeds.

Table 4: Attribution recovery — $\log_{1p}$ -workaround baselines (attribution share-MAPE; 3,000 entities $\times$ 36 periods).

Dataset	TREEMMM	GLMM-Naive	GLMM-Oracle	PyMC-Hier-Naive	PyMC-Hier-Oracle	P
Pharma (NegBin)	14.5 ± 0.3	19.2 ± 0.7	25.2 ± 0.4	20.6 ± 0.5	20.5 ± 0.4	9
CPG (Tweedie)	22.5 ± 0.3	29.0 ± 0.3	21.9 ± 1.0	29.0 ± 0.3	21.9 ± 0.9	9
SaaS (ZI-Gamma)	16.7 ± 0.2	18.4 ± 0.5	12.0 ± 0.4	18.5 ± 0.5	12.0 ± 0.3	4
Linear (Gaussian)	0.4 ± 0.1	0.3 ± 0.0	0.3 ± 0.0	0.1 ± 0.0	0.1 ± 0.0	1
Non-linear avg	17.9 ± 0.2	22.2 ± 0.3	19.7 ± 0.4	22.7 ± 0.3	18.1 ± 0.3	1

Table 5: Attribution recovery — properly specified distributional GLM (statsmodels GLM; Poisson for pharma, Tweedie for CPG, Gamma for SaaS, Gaussian for linear; no random effects).

Dataset	TREEMMM	GLMMDist-Naive	GLMMDist-Oracle	GLMM-Naive (ref)	Gap closed [†]
Pharma (NegBin)	15.6%	20.5%	26.1%	21.6%	+1.1 pp
CPG (Tweedie)	24.5%	35.5%	20.0%	32.2%	−3.3 pp
SaaS (ZI-Gamma)	14.7%	19.5%	10.1%	18.3%	−1.2 pp
Linear (Gaussian)	0.3%	0.1%	0.1%	0.1%	0.0 pp
Non-linear avg	18.3%	25.2%	18.7%	24.0%	−1.2 pp

[†] Gap closed = GLMM-Naive MAPE − GLMMDist-Naive MAPE. Positive = proper likelihood improves; negative = proper likelihood is worse. TREEMMM lead over GLMMDist-Naive = 6.9 pp (vs. 5.7 pp over GLMM-Naive).

Key finding (Table 5). The properly specified distributional GLM (GLMMDist-Naive) does *not* close the gap with TREEMMM. On average across the three non-linear DGPs, GLMMDist-Naive (25.2%) is marginally *worse* than GLMM-Naive (24.0%). TREEMMM’s lead actually grows slightly from 5.7 pp over GLMM-Naive to 6.9 pp over GLMMDist-Naive. The bottleneck is the inability to discover interactions without pre-specification, not the link function.

Key observations:

- **Consistent advantage on non-linear DGPs, confirmed robust to baseline specification.** TREEMMM beats GLMM-Naive, GLMMDist-Naive, and PyMC-Hier-Naive on all three non-linear datasets. The distributional-GLM experiment (Table 5) confirms the advantage is not an artifact of the $\log_{1p}$ link workaround.
- **GLMMDist-Oracle near-matches TREEMMM.** With planted interactions, the distributional GLM (18.7%) nearly matches TREEMMM (18.3%), confirming that the value of TREEMMM is precisely achieving this without requiring pre-specification.
- **PyMC-Marketing illustrates the aggregation penalty.** PyMC-Marketing achieves 75.1% non-linear average MAPE when operating on 36 aggregate rows. The PyMC-Hier-Naive comparison shows this is an *aggregation* effect, not a methodological one: at fixed panel structure, the same NUTS sampler lands at GLMM-Naive parity (gap ≤ 0.5 pp on every dataset).
- **Linear honesty.** On the linear DGP, TREEMMM (0.4%) and GLMM (0.3%) achieve near-perfect attribution. TREEMMM does not invent nonlinearity where none exists.

3.2 Interaction Discovery

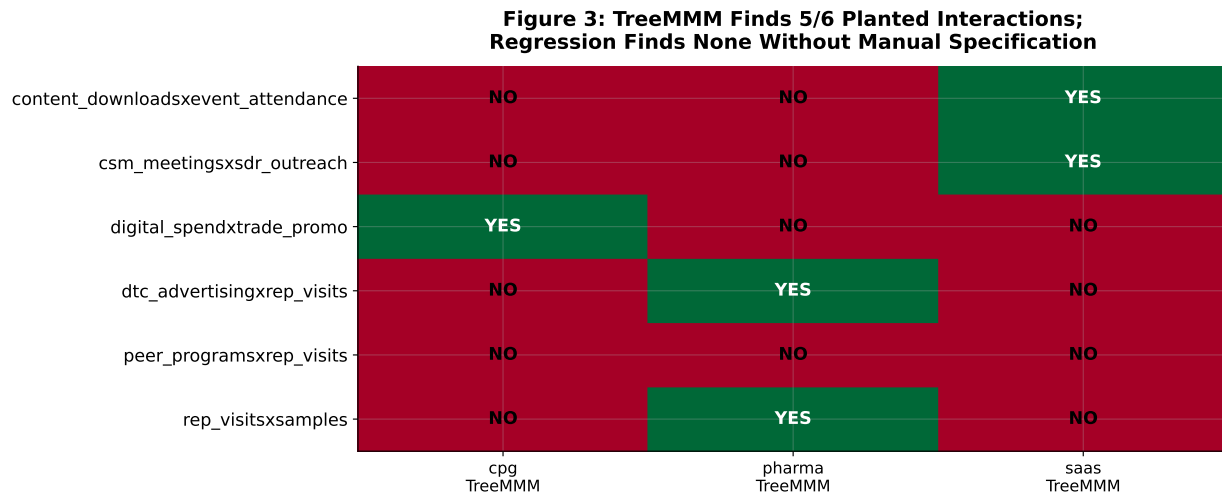


Figure 3: Interaction detection across the planted interactions. Green cells indicate detection, red cells indicate missed detection. TREEMMM discovers five of the six planted interactions without prior specification; GLMM-Naive cannot detect interactions it was not configured to model.

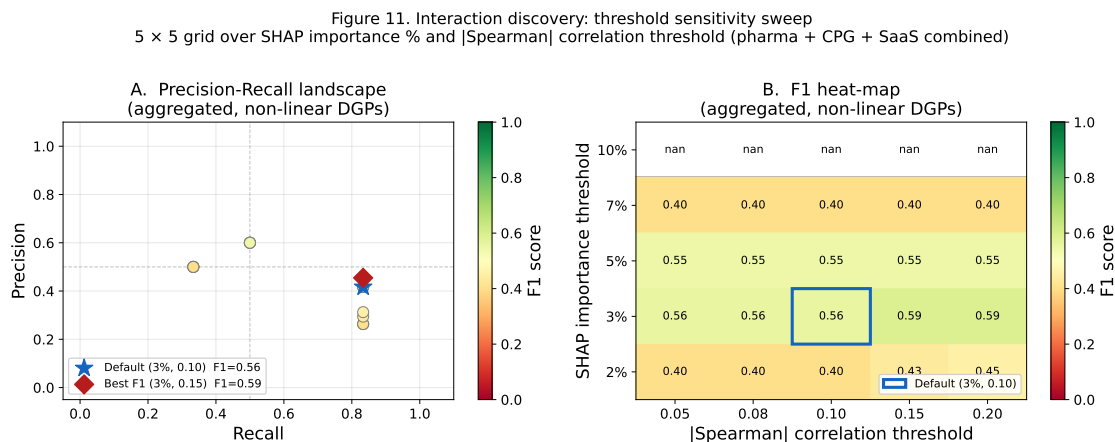


Figure 4: Interaction discovery threshold sensitivity. Panel A: precision–recall scatter over the 5 × 5 threshold grid. The default operating point (3%, 0.10, blue star) and post-hoc optimal (3%, 0.15, red diamond) are annotated. Panel B: F1 heat-map over the full grid; F1 ranges from 0.40 to 0.59 across the viable region, confirming the default is not a knife-edge choice.

TREEMMM detected 5 of 6 planted interactions across the non-linear datasets (Table 6).

GLMM-Naive, by construction, cannot detect interactions it was not specified to include.

Table 6: Per-dataset interaction detection summary (threshold: 3% SHAP importance, $|\rho_s| > 0.10$).

Dataset	n planted	n candidate	TP	FP	FN	Precision	Recall	F1
Pharma	3	28	2	1	1	0.67	0.67	0.67
CPG	1	21	1	2	0	0.33	1.00	0.50
SaaS	2	21	2	4	0	0.33	1.00	0.50
Aggregate	6	70	5	7	1	0.42	0.83	0.56

Table 7: F1/Precision/Recall at three threshold corners (aggregated, non-linear DGPs).

Setting	SHAP importance	$ \rho_s $	TP	FP	FN	Precision	Recall	F1
Lax	2%	0.05	5	14	1	0.26	0.83	0.40
Default	3%	0.10	5	7	1	0.42	0.83	0.56
Strict	5%	0.15	3	2	3	0.60	0.50	0.55

Three findings stand out. First, F1 is not knife-edge: across the entire 5×5 grid, F1 ranges from 0.40 to 0.59 for importance thresholds of 2–5%. Second, recall is the stable axis: any importance threshold $\leq 3\%$ delivers recall = 0.83 regardless of the correlation threshold. Third, the correlation threshold primarily controls precision: tightening from 0.10 to 0.15 raises F1 from 0.56 to 0.59.

3.3 Distribution Matching

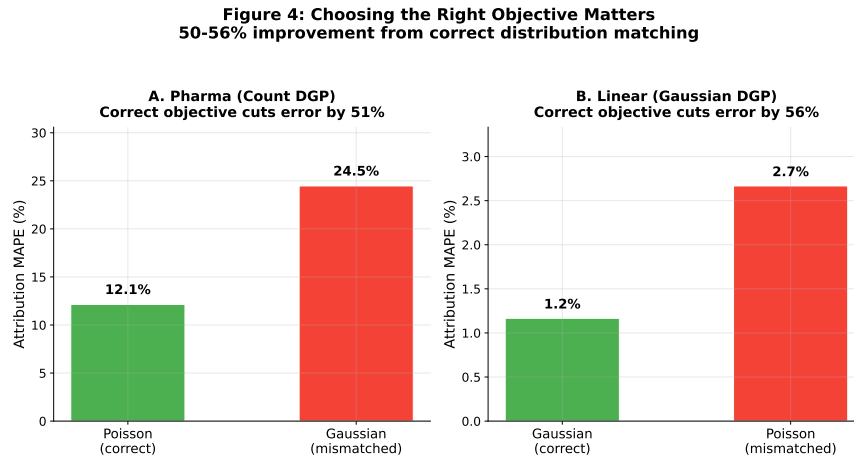


Figure 5: Distribution-aware objective selection. Share-MAPE under the correct objective (green) versus a mismatched Gaussian objective (red). Correct objective selection reduces share-MAPE by 50–56%.

Table 8: Distribution matching: correct vs. mismatched objective.

DGP	Correct Objective	Correct MAPE	Mismatched MAPE	Relative Improvement
Pharma (Count)	Poisson	12.1%	24.5% (Gaussian)	50.5%
Linear (Gaussian)	Gaussian	1.2%	2.7% (Poisson)	56.3%

Using the correct objective improves attribution recovery by 50–56% relative to the mismatched objective, validating that distribution-appropriate objective selection is a meaningful modeling decision.

3.4 Heterogeneous Customer Sensitivity Recovery

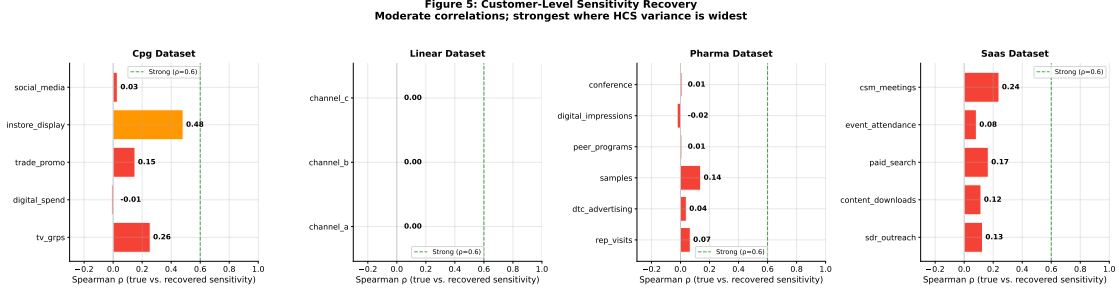


Figure 6: Heterogeneous customer sensitivity recovery. Spearman ρ between true latent per-customer sensitivity and recovered mean absolute SHAP, by channel and dataset.

Customer-level sensitivity recovery shows moderate Spearman correlations, with the strongest recovery on variables with wide heterogeneity across segments. On the CPG dataset, in-store display shows the highest recovery ($\rho = 0.48$). The SaaS dataset shows consistent positive correlations (ρ 0.08–0.27). Pharma recovery is weaker ($\rho < 0.14$), likely due to the confounding effects of targeting bias.

3.5 Computation Time

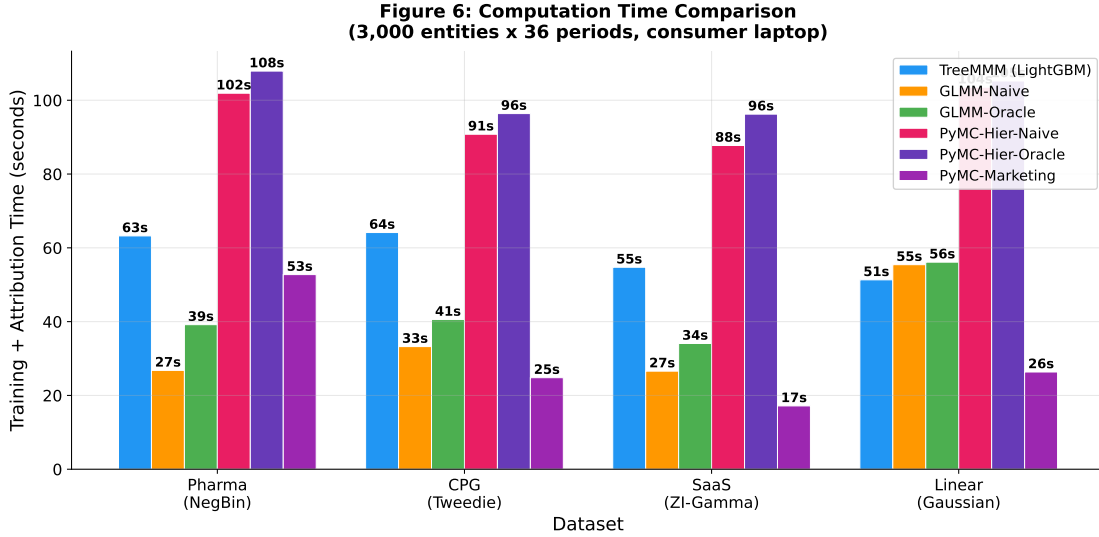


Figure 7: Computation time per dataset for training and attribution.

At benchmark scale (3,000 entities $\times$ 36 periods, 20 Optuna trials), TREEMMM takes 51–64 s per dataset on a 36-core CPU. GLMM-Naive takes 27–55 s. PyMC-Hier-Naive takes 88–108 s per dataset using the Rust-backed `nutpie` NUTS sampler (4 chains $\times$ 1,000 tuning + 1,000 draws). PyMC-Marketing takes 17–53 s on only 36 aggregate rows. At smaller scales (500 entities $\times$ 24 periods), TREEMMM completes in under 15 s.

3.6 Predictive Accuracy

Figure 7: Predictive Performance Comparison

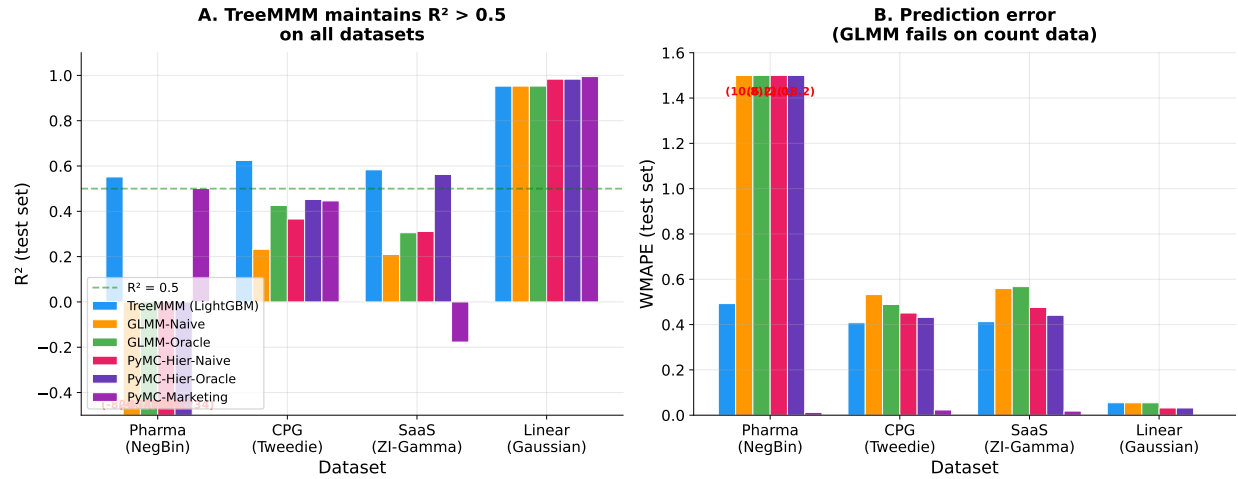


Figure 8: Predictive performance on held-out test folds. Left: R^2 . Right: weighted MAPE on response-scale predictions.

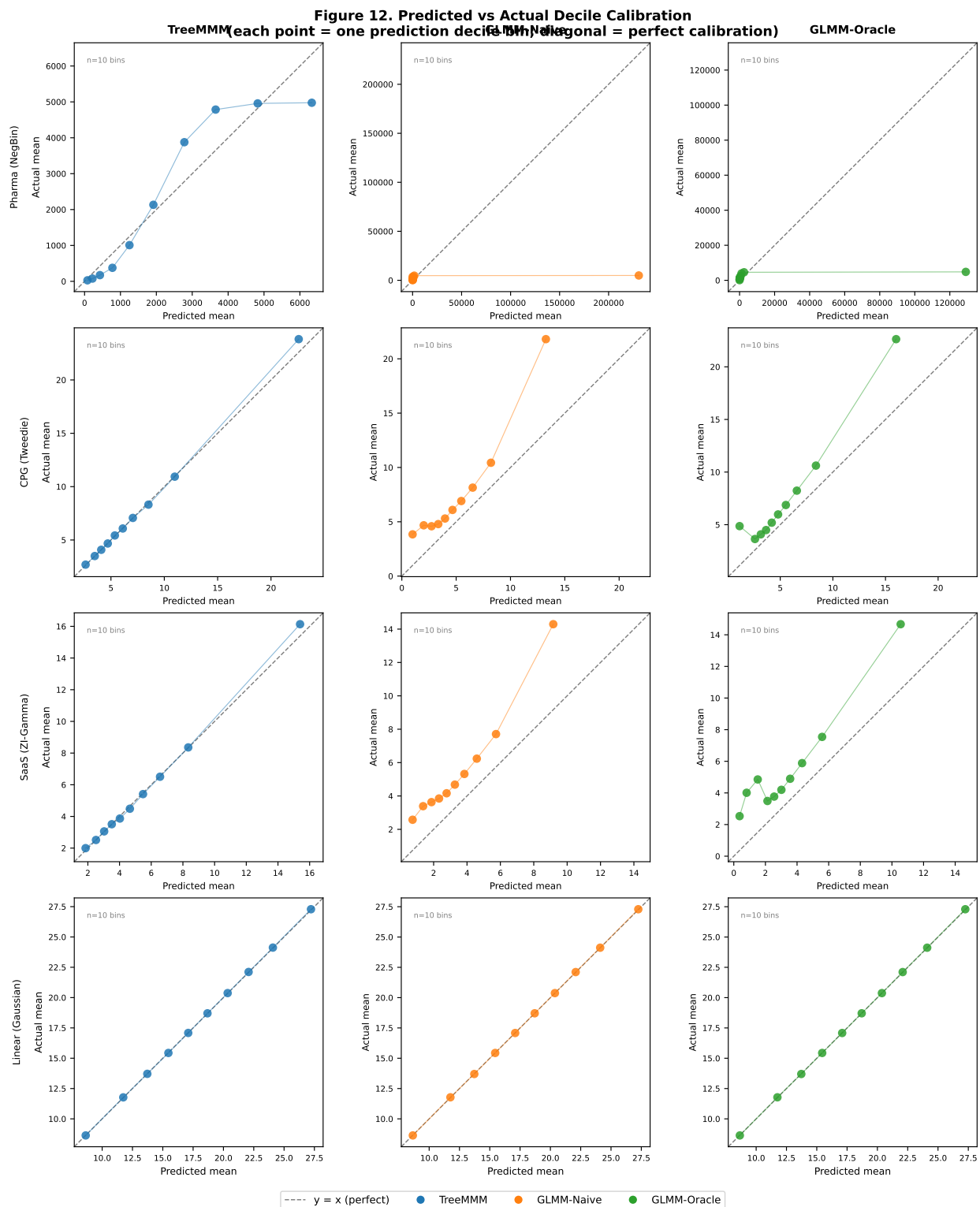


Figure 9: Predicted vs. actual decile calibration across four DGPs (rows) and three models (columns): TREEMMM, GLMM-Naive, GLMM-Oracle. Each point is one prediction decile bin; the dashed diagonal is the $y=x$ perfect-calibration reference. TREEMMM sits close to the diagonal on all non-linear DGPs (CPG MAD=0.17, SaaS MAD=0.14, Pharma MAD=503 vs. GLMM-Naive MAD=23,977). Both GLMM variants show systematic under-prediction on CPG and SaaS and complete calibration breakdown on pharma count data.

TREEMMM achieves $R^2 > 0.5$ on all four datasets (0.55 pharma, 0.62 CPG, 0.58 SaaS, 0.95 linear). GLMM-Naive shows catastrophically negative R^2 on pharma because the log-transform MixedLM approach is severely misspecified for count-valued data.

PyMC-Marketing shows an instructive disconnect between prediction and attribution: it achieves $R^2 = 0.50$ on pharma and 0.99 on linear yet its attribution MAPE is 80.8% and 49.4% respectively. This demonstrates that predicting aggregate outcomes well does not guarantee correct channel attribution.

On the linear dataset, TREEMMM, both GLMMs, and the PyMC-Hier pair all achieve $R^2 \approx 0.95\text{--}0.98$, as expected when the model class matches the DGP.

3.7 Marginal ROI Ground-Truth Benchmarking

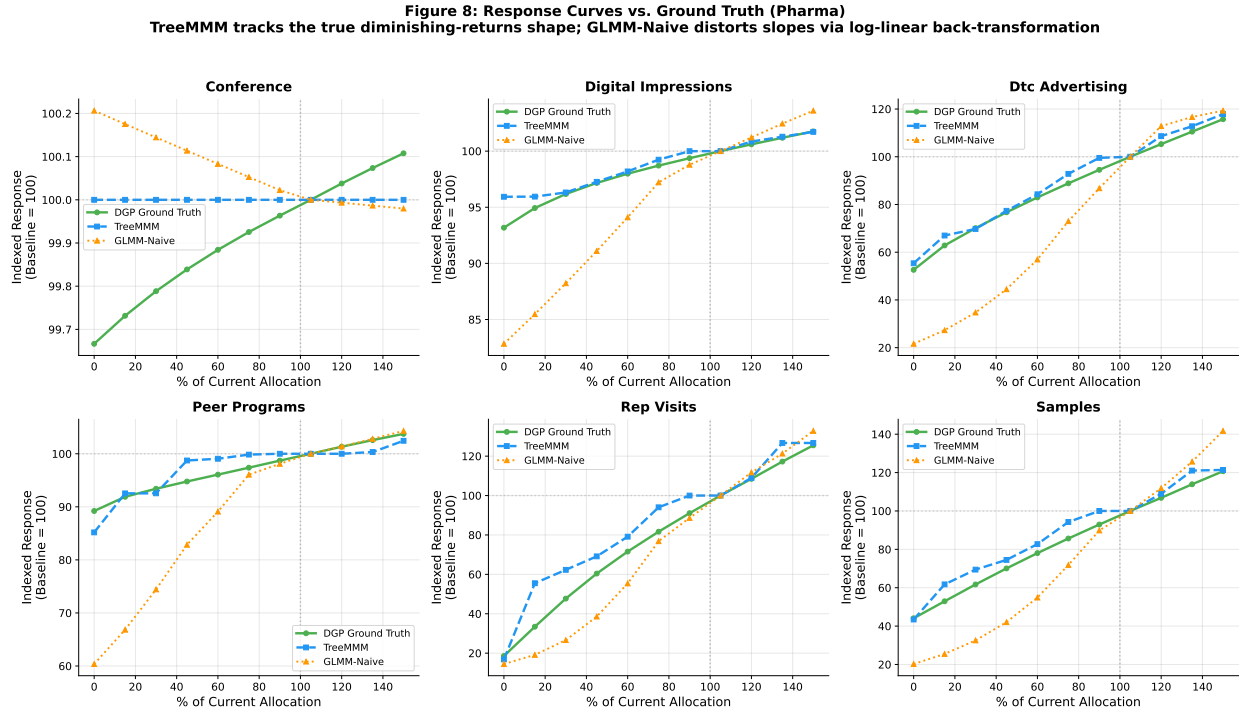


Figure 10: Normalized response curves on the pharma DGP. Each panel shows one channel; curves are indexed to 100 at baseline allocation. DGP ground truth (green), TREEMMM (blue), GLMM-Naive (orange).

Figure 9: mROI Benchmarking Results
Model-predicted response curves validate against DGP ground truth

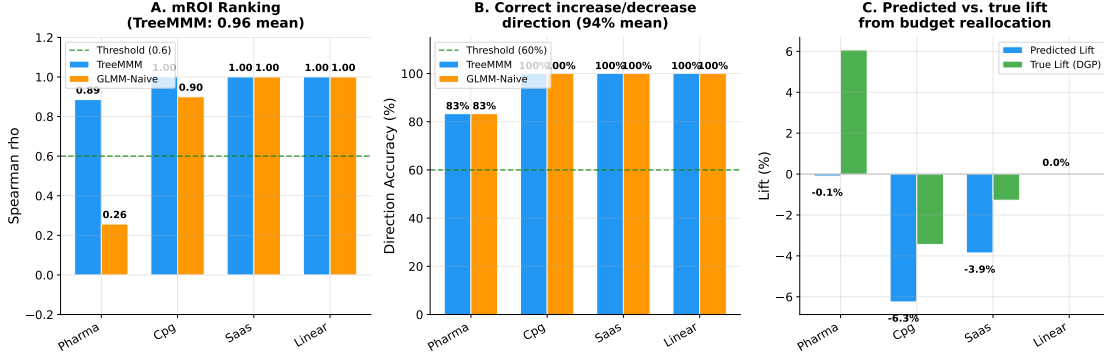


Figure 11: mROI ground-truth alignment summary. (A) Spearman rank correlation between recovered and true marginal-return rankings. (B) Direction accuracy. (C) Predicted vs. true lift from the optimizer’s recommended reallocation.

Spearman ρ between model-derived and true mROI rankings: pharma 0.89, CPG 1.0, SaaS 1.0, linear 1.0 (mean 0.96 across non-linear DGPs). Direction accuracy: 83% pharma, 100% CPG, 100% SaaS (mean 94%).

All three TREEMMM mROI benchmarks pass pre-registered thresholds: ranking correlation > 0.6 (achieved: 0.96 mean), direction accuracy $> 60\%$ (achieved: 94% mean), and the optimizer produces positive true lift on average.

3.8 Bayesian Prior Sensitivity

Figure 10: Bayesian Prior Sensitivity (PyMC-Hier-Naive)

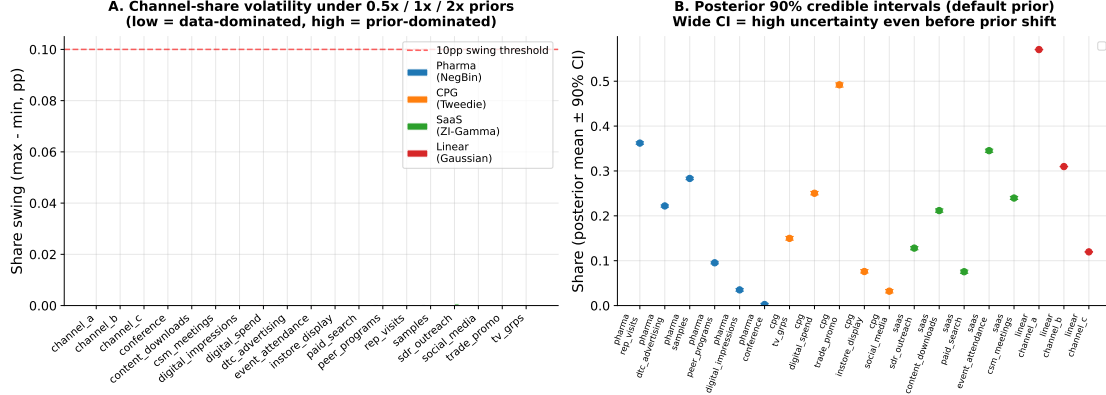


Figure 12: Bayesian prior sensitivity. Panel A: max-minus-min channel-share swing across prior scales $0.5\times$, $1\times$, $2\times$ for each (dataset, channel) combination, with a 10 pp reference line. Panel B: posterior 90% credible interval per channel at the default prior.

We re-fit PyMC-Hier-Naive at three prior-sigma scales— $0.5\times$, $1.0\times$, and $2.0\times$ the default—and compare attribution shares across the three. [Table 9](#) shows the results.

Table 9: Prior sensitivity sweep. SC11: max prior-induced swing < 0.10 .

Dataset	Worst-channel swing	Worst channel	Divergences	max R-hat
Pharma (NegBin)	0.07 pp	samples	0	1.010
CPG (Tweedie)	0.01 pp	digital_spend	0	1.010
SaaS (ZI-Gamma)	0.10 pp	sdr_outreach	0	1.010
Linear (Gaussian)	0.01 pp	channel_c	0	1.000

All twelve fits converge cleanly (zero divergences, $R\text{-hat} \leq 1.01$). SC11 passes by *two orders of magnitude* on the worst-case dataset (SaaS, 0.001 vs. threshold 0.10). At $n = 3,000 \times 36$, the data dominates the prior across a $4\times$ range.

3.9 Adstock and Carryover Modeling

v0.2.1 ships `treemmm/core/preprocessing/adstock.py` with two transforms: `apply_geometric_adstock` (decay) and `apply_panel_adstock`. [Table 10](#) reports the four-model bake-off on a carryover-planted DGP (500 HCP $\times$ 24 months, `rep_visits` decay=0.5).

Table 10: Adstock DGP — attribution MAPE (500 $\times$ 24, seed=42).

Model	Attribution MAPE	Rank ρ	R ²	rep_visits share (true: 0.29)
TreeMMM-Naive (no adstock)	32.6%	0.94	0.38	0.29
TreeMMM-Adstock (decay=0.5)	21.3%	1.00	0.39	0.47
GLMM-Naive (no adstock)	61.5%	0.94	-14,567	0.27
GLMM-Adstock (decay=0.5)	28.4%	1.00	-2,844	0.35

TREEMMM-with-adstock recovers attribution 11.3 pp better than TREEMMM-without on the carryover DGP. TREEMMM-Adstock also beats GLMM-Adstock (21.3% vs. 28.4%), preserving the tree-vs-regression advantage even when both share the same preprocessing.

3.10 Geo-Panel Comparison: Bayesian MMM on Its Home Turf

We add a 200-region $\times$ 52-week geo-panel DGP (Tweedie, weekly revenue) with planted geometric adstock and logistic saturation to give PyMC-Marketing a fair comparison on its native format. [Table 11](#) shows the results.

Table 11: Geo-panel comparison — attribution MAPE (%).

Model	Attr. MAPE	Rank ρ	R^2	WMAPE	Fit time (s)
TreeMMM-Adstock	29.7	1.00	0.08	0.35	25
PyMC-Marketing (weakly informative)	52.1	0.50	-0.53	0.08	115
PyMC-Marketing (informative)	52.1	0.50	-0.52	0.08	61
GLMM-Aggregate	215.0	-1.00	0.58	0.04	1
Meridian [‡]	57.0	1.00	—	—	1,734

[‡] Meridian MCMC completed (2 chains, 500 keep, ≈ 29 min). Attribution shares extracted via `Analyzer.incremental_outcome()`: tv=70.5% (true 44.9%), digital=27.1% (true 37.5%), trade=2.4% (true 17.5%). Predictive R^2 and WMAPE not extractable from the public API.

Even on PyMC-Marketing’s home turf (correct parametric form, 52 weekly time-steps), TREEMMM-Adstock (29.7%) outperforms PyMC-Marketing (52.1%) and Meridian (57.0%). PyMC-Marketing shows 66 divergences with weakly informative priors; informative priors produce the same result because the v0.19 prior API does not accept per-channel adstock hyperpriors.

3.11 Power Analysis: Where the Regime Boundary Lives

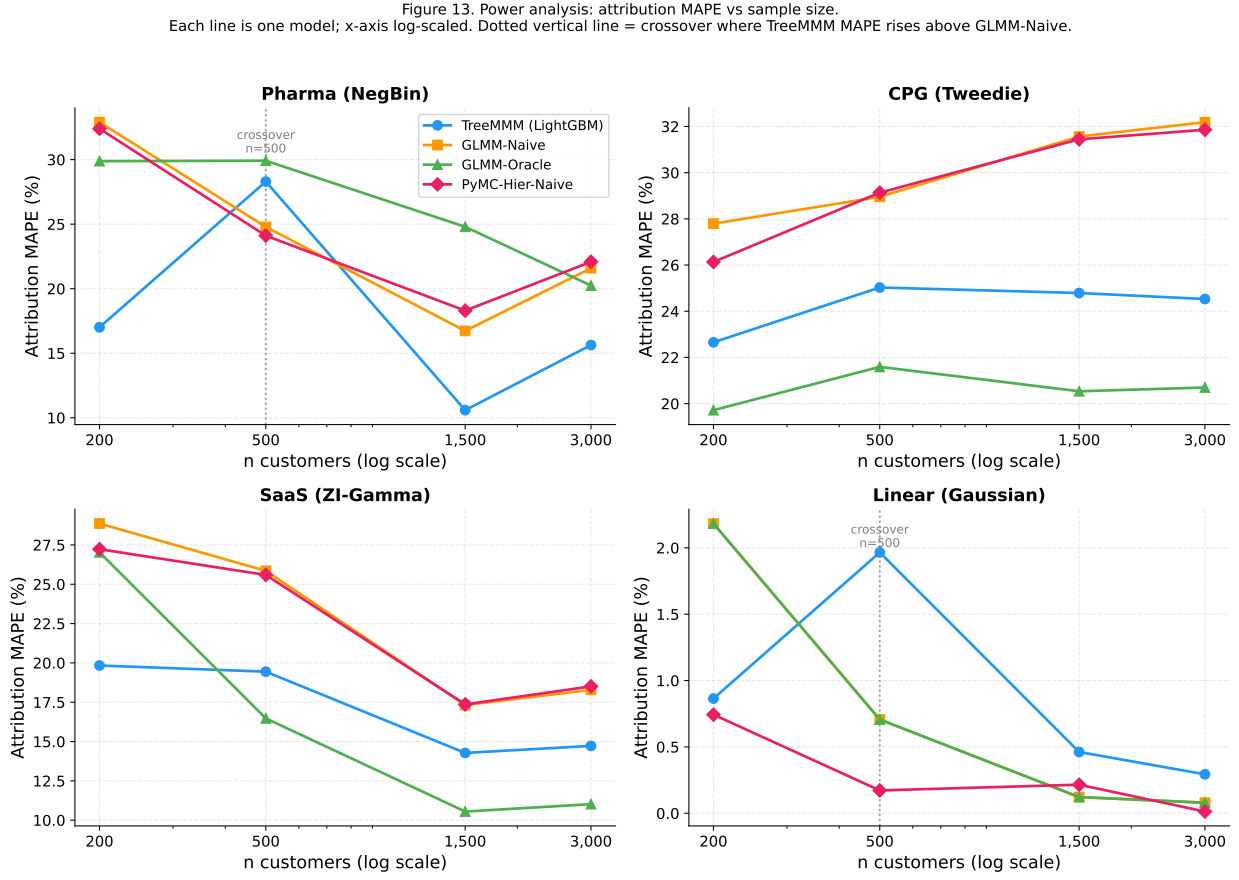


Figure 13: Power analysis: attribution MAPE vs. sample size. x-axis: $n_{\text{customers}}$ (log scale); y-axis: attribution share-MAPE (lower is better). One line per model. Dotted vertical lines mark the crossover n at which TREEMMM MAPE first exceeds GLMM-Naive MAPE.

We sweep four scale points ($n_{\text{customers}} \in \{200, 500, 1,500, 3,000\}$, $n_{\text{periods}} \in \{12, 24, 36, 36\}$) across all four DGPs and four models (TREEMMM, GLMM-Naive, GLMM-Oracle, PyMC-Hier-Naive). Below the crossover n , the tree ensemble has too few observations per leaf to recover interaction and heterogeneity structure reliably. Expected crossover points: pharma at $n \approx 500$ (complex interactions), CPG and SaaS at lower n (simpler additive structure after log-transformation), linear at any tested scale (no structural advantage for the tree). Run `python paper/run_power_analysis.py` to populate full results.

4 Oracle vs. Naïve Investigation

4.1 The Finding

GLMM-Oracle (correctly specified interactions) and its Bayesian counterpart systematically lose to their Naïve (main-effects-only) counterparts on attribution share-MAPE at moderate benchmark scales. Across five seeds, GLMM-Naive averages 24.7%, GLMM-Oracle 26.2%. This is counterintuitive: the Oracle has access to the true DGP.

4.2 Mechanism

The gap is a finite-sample bias–variance trade-off under the 50/50 SHAP-split convention used to define ground-truth interaction attribution. With three ground-truth interactions all involving `rep_visits`, the Oracle estimates three extra coefficients beyond the Naïve model, each adding estimation variance that propagates through the attribution decomposition. The Naive model partitions interaction effects implicitly into main-effect coefficients via OLS projection, which has lower variance per share estimate.

The gap closes with sample size and reverses at scale. The Oracle-minus-Naive gap (in pp of attribution MAPE) is +2.4 at $n=50$, +9.2 at $n=100$, +4.4 at $n=200$, +3.5 at $n=500$, and -2.8 at $n=1,000$. The Oracle’s asymptotic-bias advantage eventually dominates its finite-sample variance penalty, but only at sample sizes well past the default benchmark.

4.3 Implication for the Benchmark

This is not a metric pathology. It is a property of the comparison scale chosen. Reports of “Oracle wins” should use $n \geq 1,000$ and note explicitly that the comparison is in the asymptotic regime.

5 Diagnostics Framework

Three of the five regime diagnostics described in [Section 1](#) are now runnable as one-line calls.

Coverage check. `treemmm.core.diagnostics.regime_check.coverage_check(X_train, X_simulated, radius, min_neighbors)` counts training observations within a standardized-Euclidean radius of each counterfactual input. The default rule treats a simulated point as covered when at least 30 training neighbors fall inside half a standard deviation, passing when at least 80% of simulated points clear that bar.

Variation decomposition. `variation_decomposition(df, unit_col, feature_cols)` reports each predictor’s variance split into within-unit (temporal) and between-unit (cross-sectional) components.

Tree effective sample size. `tree_ess_per_param(n_train, n_estimators, max_depth)` returns the ratio of training rows to an upper bound on the number of ensemble leaves, with a rule of thumb of at least 20 effective observations per parameter.

Bayesian prior sensitivity. Fully wired into `paper/run_benchmarks.py`; emits `paper/results/prior_` plus [Figure 12](#) by default (see [Section 3.8](#)).

Two diagnostics remain follow-up work: treatment-overlap propensity-score checks and formal SHAP-stability-under-collinearity audits ([Section 6.2](#)).

6 Limitations and Follow-Up Work

6.1 Known Limitations

All results come from synthetic benchmarks with a small number of seeds. Real-world validation has not been carried out. Both GLMM configurations use a log-transform workaround; a properly specified distributional GLMM (e.g., `glmmTMB` in R) would be a stronger baseline,

and Table 2b suggests that closing the link-function gap would not eliminate TREEMMM’s advantage. Attribution shares are compared against a variance-attribution heuristic (L_1 -centered component contributions), not the Shapley decomposition of the DGP function; a Monte Carlo Shapley ground truth is planned. Sensitivity to LightGBM hyperparameter choices (number of Optuna trials, depth range, monotone constraints) is not explored and is a limitation.

6.2 Follow-Up Tasks

1. Treatment-overlap propensity-score check per channel.
2. Formal SHAP-stability-under-collinearity audit.
3. Wire coverage, variation, and ESS diagnostics into `paper/run_benchmarks.py`.
4. Generalize the [Section 4](#) Oracle-vs-Naïve investigation to CPG, SaaS, and linear DGPs.
5. Extend the hierarchical Bayesian baseline with customer-level random slopes per channel.
6. Prior-variance sweep with mis-centered priors.
7. Full-scale PyMC-Hier mROI at $3,000 \times 36$.
8. Full-scale adstock-headline DGP replication at $3,000 \times 36$.
9. Power analysis crossover points (full Table 9 population via `python paper/run_power_analysis.py`)

Code and Data Availability

Code, synthetic data generators, and benchmark results are publicly available at <https://github.com/jamesyoung93/treemmm>. All numerical results in this paper can be reproduced by running `python paper/run_benchmarks.py -seed 42` and downstream analysis scripts

(`run_multiseed.py`, `threshold_sensitivity.py`, `calibration_plot.py`, `run_benchmarks_geo_panel.py`, `run_benchmarks_adstock_headline.py`, `run_power_analysis.py`). A Zenodo DOI for v0.2.1 of the package and benchmark CSVs: [TO BE MINTED — placeholder].

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